

Name: _____

1. Use the table below to answer the following questions

	P_c	Q_c	P_f	Q_f
2021	1000	10	4	500
2022	10	1200	5	510

- (a) Calculate the nominal GDP for the years 2021 and 2022.
- (b) Calculate the real GDP for the year 2022, using the year 2021 as the base year.
- (c) Calculate the real GDP for the year 2021, using the year 2022 as the base year.
- (d) Calculate the real GDP growth using the prices from the year 2021.
- (e) Calculate the real GDP growth using the prices from the year 2022.
- (f) What accounts for the substantial disparity between the two growth rates?

Solution:

(a) Nominal GDP for year 2021:

$$GDP_{21} = 1000 \times 10 + 4 \times 500 = 12000$$

Nominal GDP for year 2022:

$$GDP_{22} = 10 \times 1200 + 5 \times 510 = 14550$$

(b) The real GDP for year 2022 is

$$GDP'_{22} = 1000 \times 1200 + 4 \times 510 = 1,202,040$$

(c) The real GDP for year 2021 is

$$GDP'_{21} = 10 \times 10 + 5 \times 500 = 2600$$

(d) The growth is $g_Y = 99.17 = 9917\%$

(e) The growth is $g_Y = 4.596 = 459.6\%$

(f) The reason Laspeyres (part d) and Paasche (part e) Price Indexes produce differing results is rooted in their use of the prices. In both calculations, prices are fixed, at different points in time — one fixed at initial prices (2021), while the other is fixed at end-of-period prices (2022). Consequently, the Laspeyres Index typically underestimate the actual impact of price shifts, while the Paasche Index tends to overstate this impact.

In our specific scenario, consider the case of good C, where prices increased from 1000 to 10, while simultaneously witnessing a surge in quantity from 10 to 1200, indicating increased

demand due to the reduced price (law of demand). When employing the Laspeyres Index, the price remains at 1000, guided by the assumption that consumers exhibit no changes in their behavior in response to shifting prices.

Alternatively, you can think of the price as the weight for the quantity of each good. With the Laspeyres or Paasche Index way of calculating the real GDP, the weight are indeed fixed. Therefore, in the part d), the weight on the good C is remain very high, even though the market had experienced a large downward movement along the demand curve (assume only the supply curve is moving).

2. Consider in an economy with two consumption goods and two investment goods, use the table and answer the question below:

	P_{C1}	Q_{C1}	P_{C2}	Q_{C2}	P_{I1}	Q_{I1}	P_{I2}	Q_{I2}
2012	11	110	18	50	22	11	52	11
2013	12	130	22	52	23	12	47	15

- (a) Calculate the real consumption, real investment and real GDP, using the price in 2012.
- (b) Calculate the real consumption, real investment and real GDP, using the price in 2013.
- (c) Calculate the Laspeyres, Paasche, and Fisher price indices for the change in real consumption, real investment, and real GDP from 2012 to 2013.
- (d) Calculate Real consumption, Real investment and Real GDP in chained prices, benchmarked to 2013.
- (e) Sum the real consumption and real investment and compare it to the real GDP.

Solution:

- (a) The real consumption would be $11 \times 110 + 18 \times 50 = 2110$ in year 2012 and $11 \times 130 + 18 \times 52 = 2366$ in year 2013

The real investment would be $22 \times 11 + 52 \times 11 = 814$ in year 2012 and $22 \times 12 + 52 \times 15 = 1044$ in year 2013

The real GDP are 2924 and 3410

- (b) The real consumption would be $12 \times 110 + 22 \times 50 = 2420$ in year 2012 and $12 \times 130 + 22 \times 52 = 2704$ in year 2013

The real investment would be $23 \times 11 + 47 \times 11 = 770$ in year 2012 and $23 \times 12 + 47 \times 15 = 981$ in year 2013

The real GDP are 3190 and 3685

- (c) Laspeyres: $L_C = \frac{2366}{2110} = 1.121$ $L_I = \frac{1044}{814} = 1.283$ and $L_Y = \frac{3410}{2924} = 1.166$

Paasche: $Pi_C = \frac{2704}{2420} = 1.117$ $Pi_I = \frac{981}{770} = 1.274$ and $Pi_Y = \frac{3685}{3190} = 1.155$

Fisher: $F_F = \sqrt{1.121 \times 1.117} = 1.119$, $F_I = \sqrt{1.283 \times 1.274} = 1.278$, $F_Y = \sqrt{1.166 \times 1.155} = 1.161$

- (d) The real consumption, real investment and real GDP in 2013 at 2013 prices are 2704, 981 and 3685. To calculate the chained weight value, we use the following formula

$$x_{t-1} G_t = x_t \rightarrow x_{t-1} = \frac{x_t}{G_t}$$

The G_t here is the gross growth rate, which is the same as our index calculation as we did not multiply it by 100.

$$C_{2012} = \frac{C_{2013}}{G_{2013}} = \frac{2704}{1.119} = 2415.7$$

$$I_{2012} = \frac{I_{2013}}{G_{2013}} = \frac{981}{1.274} = 767$$

$$Y_{2012} = \frac{Y_{2012}}{G_{2013}} = \frac{3685}{1.161} = 3174.9$$

- (e) Summing the C_{2012} and I_{2012} , we have $2415.7 + 767 = 3183.15$. Compare this number to the real GDP in chained weight, $3183.15 > 3174.9$. It is because the price of consumption and investment grow at a different rate.